

UNIT 3

EMERGING COMPUTING TECHNOLOGIES: BEYOND SUPERCOMPUTERS



Learning Objectives

1. Understand the advancements in computing technologies beyond supercomputers.
2. Define and explain the concepts of quantum computing, grid computing, cloud computing, edge computing, and AI-optimized computers.
3. Identify the applications of these technologies in real life.
4. Appreciate the role of modern computing in solving complex global challenges.

Computers have transformed the way we live, learn, and work. After the invention of supercomputers, which are capable of performing trillions of calculations in seconds, scientists and engineers continued to push the boundaries of what computers can achieve. New computing technologies like quantum computing, grid computing, cloud computing, edge computing, and AI-optimized computing have emerged. These advanced technologies are designed to solve even more complex problems efficiently and support the growing needs of modern life. Let us explore these exciting innovations.

1. Quantum Computing

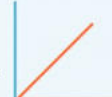
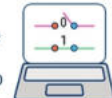
Quantum computing is a revolutionary technology that uses the principles of quantum mechanics to process information. Unlike traditional computers that use bits (0s and 1s), quantum computers use qubits, which can represent both 0 and 1 at the same time.

Key Features:

- Performs calculations at incredible speeds.
- Solves problems that are impossible for traditional computers.

Applications:

- Cryptography (secure communication)
- Drug discovery (designing medicines faster)
- Climate modelling (predicting weather and climate changes)

Quantum Computing	Vs.	Classical Computing
 Calculates with qubits, which can represent 0 and 1 at the same time		 Calculates with transistors, which can represent either 0 or 1
 Power increases exponentially in proportion to the number of qubits		 Power increases in a 1:1 relationship with the number of transistors
 Quantum computers have high error rates and need to be kept ultracold		 Classical computers have low error rates and can operate at room temp
 Well suited for tasks like optimization problems, data analysis, and simulations		 Most everyday processing is best handled by classical computers

2. Grid Computing

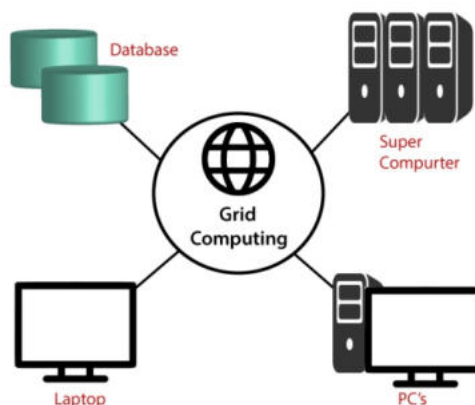
Grid computing connects multiple computers over a network to work together as a single system. These computers share resources to solve large-scale problems.

Key Features:

- Utilizes the unused processing power of many computers.
- Performs high-performance computing tasks.

Applications:

- Scientific research (analyzing large dataset)
- Simulations (predicting natural disasters)



3. Cloud Computing

Cloud computing allows users to access and store data and programs on the internet instead of using their own computers or servers. It provides resources on demand, such as storage, servers, and applications.

Key Features:

- Accessible from anywhere with an internet connection.
- Reduces the need for expensive hardware.

Applications:

- Online services like Google Drive and Dropbox
- Business operations (managing large amounts of data)
- E-learning platforms



4. Edge Computing

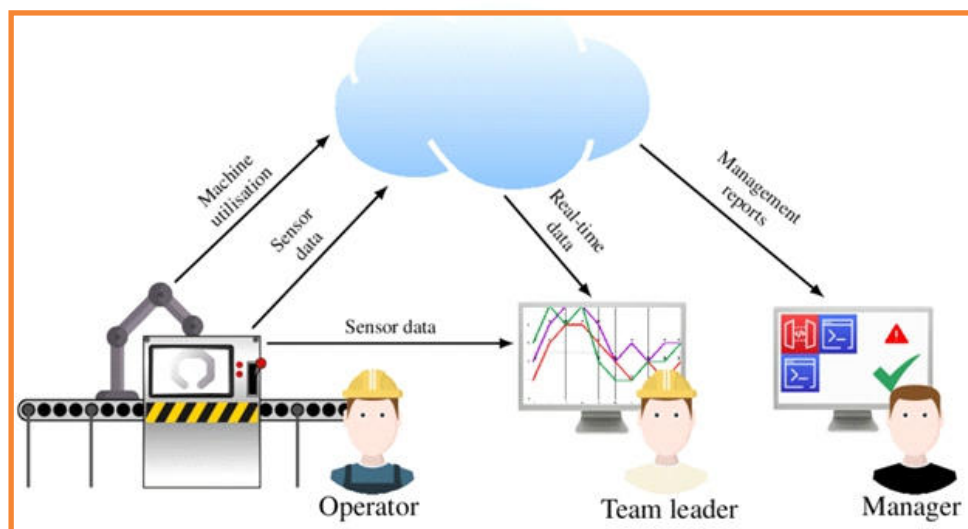
Edge computing processes data close to its source, such as on Internet of Things (IoT) devices, instead of sending it to a central server. This reduces delays and improves speed.

Key Features:

- Processes data in real-time.
- Reduces internet bandwidth usage.

Applications:

- Autonomous vehicles (real-time decision-making)
- Smart cities (managing traffic and energy)
- Healthcare devices (monitoring patients remotely)



A diagram of an IoT device processing data locally.

5. AI-Optimized Computing

AI-optimized computers are specially designed for artificial intelligence tasks. They use advanced processors like GPUs (Graphics Processing Units) and TPUs (Tensor Processing Units) to perform tasks faster.

Key Features:

- Optimized for machine learning and deep learning.
- Processes large amounts of data simultaneously.

Applications:

- Self-driving cars
- Facial recognition systems
- Virtual assistants (like Alexa and Siri)



Emerging computing technologies have revolutionized how we solve complex problems and manage data. From quantum computing to AI-optimized machines, these advancements continue to shape the future, offering new possibilities for innovation in science, business, healthcare, and more.

EXERCISES

01: Choose the correct answer.

1. What type of computer uses qubits instead of bits?

- a) Supercomputer
- b) Cloud computer
- c) Quantum computer
- d) AI-optimized computer

2. Which computing technology connects multiple computers to solve large problems?

- a) Quantum computing
- b) Grid computing
- c) Edge computing
- d) Cloud computing

3. What is the main feature of cloud computing?

- a) Stores data locally
- b) Reduces delays
- c) Provides on-demand internet-based resources
- d) Performs AI tasks

4. Which technology is essential for autonomous vehicles?

- a) AI-optimized computers
 - b) Edge computing
 - c) Cloud computing
 - d) Grid computing
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5. AI-optimized computers are mainly used for:

- a) Storing large amounts of data
- b) Real-time traffic management
- c) Training machine learning models
- d) Analyzing weather patterns

6. Which computing technology is primarily used in smart cities for traffic management?

- a) Quantum computing
- b) Cloud computing
- c) Edge computing
- d) Grid computing

7. What is the key advantage of grid computing?

- a) Solving small-scale problems efficiently
- b) Using unused processing power across multiple computers
- c) Performing tasks locally
- d) Storing data in the cloud

8. What is a major application of quantum computing?

- a) Managing smart home devices
 - b) Weather forecasting
 - c) Cryptography
 - d) Online learning platforms
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9. Which type of computer processes data closest to its source?

- a) AI-optimized computer
- b) Edge computer
- c) Quantum computer
- d) Cloud computer

10. What type of processor is typically used in AI-optimized computers?

- a) CPUs
- b) GPUs and TPUs
- c) Quantum processors
- d) Analog processors

02: Fill in the blanks.

1. _____ computing processes data close to its source, such as on IoT devices.
2. _____ computers use qubits to perform extremely fast calculations.
3. _____ computing provides internet-based resources like storage and applications.
4. AI-optimized computers use advanced processors like _____ and TPUs.
5. _____ computing connects multiple computers to work together as a single system.

03: Select True or False for each statement.

	TRUE	FALSE
Quantum computers use binary bits for processing.	☐	☐
Cloud computing requires expensive hardware for users.	☐	☐
Edge computing reduces internet bandwidth usage.	☐	☐
Grid computing involves only one computer solving problems.	☐	☐
AI-optimized computers are not used in virtual assistants.	☐	☐

04: Answer the following questions briefly.

1. What are qubits, and how are they used in quantum computers?
2. How does cloud computing benefit users?
3. Describe an application of edge computing in real life.
4. What makes AI-optimized computers suitable for machine learning tasks?
5. Explain how grid computing distributes tasks across multiple computers.
6. What role do embedded computers play in home appliances?

05: Provide detailed answer for the following questions.

1. Explain quantum computing and its key differences from traditional computing. Include its applications.
2. Describe the concept and advantages of cloud computing. How is it beneficial in education and business?
3. Discuss the role of edge computing in real-time applications like autonomous vehicles and smart cities.
4. Compare and contrast grid computing and AI-optimized computers, highlighting their unique features and applications.

Lab Activity 1: Watching and Discussing a Quantum Computing Video



Objective: Help students understand quantum computing concepts through a visual and auditory medium.

Steps:

1.Watch the Video:

- Play a short, engaging video on quantum computing basics, such as:

[Quantum Computing for Beginners](#) (3–5 minutes).

2.Class Discussion:

- Ask the students:
- What did you find most interesting about quantum computing?
- How is quantum computing different from traditional computing?
- Can you think of ways quantum computing might solve real-world problems?

Lab Activity 2: Understanding Quantum Bits (Qubits)



1. Research and Write:

Open MS Word and create a document titled

"What are Qubits?"

- Write a short paragraph about qubits, explaining how they differ from traditional bits.
- Include an example or analogy to make the concept relatable (e.g., comparing a spinning coin to qubits vs. a stationary coin for bits).

2. Create a Visual Representation:

- Use MS Word's **Shapes and SmartArt** tools to draw a simple diagram comparing bits (0s and 1s) with qubits (0, 1, and both).
- Add labels and a brief explanation below the diagram.

3. Real-World Applications:

- Research one application of quantum computing (e.g., cryptography, drug discovery, or climate modelling).
- Create a table in MS Word with the following headings:
 - Application Name
 - How Quantum Computing Helps

- Why It's Important
- Fill in the table with at least three examples.

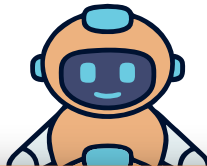
4. Critical Thinking:

- Write a short paragraph answering:
- Why do you think quantum computing is important for the future?
- Can quantum computing impact daily life? How?

5. Submission:

Save the file as "**QuantumComputing-LabActivity docx**" and submit it to your teacher.

Lab Activity 3: Exploring Cloud Storage Services



- **Objective:** Learn how to use cloud computing services and understand how cloud computing simplifies file management and collaboration.
- **Task:**
 - Create an account on a cloud storage service like Google Drive or Dropbox.
 - Upload, download, and share files.
 - Observe how cloud computing provides easy access to files from different devices.

Lab Activity :4 Organizing Files in Cloud Storage



Objective: Understand how to organize and manage files effectively using cloud storage services.

Steps:

1.Create Folders:

- Log in to your cloud storage account (e.g., Google Drive or Dropbox).
- Create separate folders for schoolwork, personal documents, and multimedia (images/videos).

2. Organize Files:

- Upload at least five files (e.g., documents, images, or videos) and sort them into the appropriate folders.

3. Rename Files and Folders:

- Rename the uploaded files and folders with meaningful names to improve organization.

4. Reflection Task:

- Write a short paragraph in MS Word about how cloud storage helps in file organization and management.
- Save and upload the Word document to the same cloud folder.

Lab Activity :5 Collaborating on a Shared Document



Objective: Explore how cloud storage services enable real-time collaboration on documents.

Steps:

1. Create a Shared Document:

- Log in to a cloud storage account (e.g., Google Drive).
- Create a new Google Doc or upload an existing Word document.

2. Share the Document:

- Share the document with your teacher or classmates by adding their email addresses.

3. Collaborate in Real Time:

- Work together with the shared participants to edit or comment on the document simultaneously.

4. Observation Task:

- Note how multiple users can access and edit the same document at the same time.
- Discuss how this simplifies group projects and collaborative tasks.